

1. Introducing the topic

Today I will give you a presentation about smart home. First of all, I chose this topic because it's an interesting form of new technology and I also want to work with IoT in the future, however I will come back this point later.

I have divided my presentation into 7 parts:

First, I would like to start with the concept.

Second, I will focus on the technical solutions.

After that, we will deal with the network and components.

Then, I will move on to systems and technologies.

Finally, the last section will be about diagrams.

During the presentation don't interrupt me, please! It can take my attention off the speech. If you don't mind, we will leave the questions till then end.

2. What is a smart home?

Now, let's begin with some questions. Could there be something smarter than your phone? What is a smart home accurately?

Yes, many things can be smarter than your phone. For example, your washing machine, your bed, but even your house.

A smart home is a household with controllable electronic devices, which make everyday tasks easier.

3. Technical solutions

This leads me to my second point: Technical solutions.

Although this concept appears to be involved only IT intervention, the implementation requires the cooperation of many specialized fields.

Including electronics, architecture, automation, telecommunications etc.

In order to build a smart home, much more special intervention is required than for an average house.

I mentioned earlier some smart devices. A really specialized field of computer science deals with these. In fact there is a professional term for this field:

IoT, in other words Internet of Things. Officially, it includes physical objects with sensors, processing ability, software and other technologies that connect and exchange data with other devices and systems over the Internet.

4. Network and components

Let's turn our attention to the next slide. In the first place, it's necessary to select a network solution for our devices. There are two options and plus one. We can utilize radio frequency RFID, or WI-FI technology. On the other hand, the best solution is a combination of the two.

However, it demands components for the operation of the system:

A sensor is required to determine the temperature.

It can't operate without executive functions, which control the switches.

Next things are buses. Now I'm talking about buses in the IT sense of course. The bus enables the transmission of data and supply voltages.

At the same time, the system also includes a communication interface to establish a connection between computer and human.

5. System and technologies

Now let's move on to the next and most interesting part. What can a smart home offer us?

Let me enumerate, please:

Automatically adjustable brightness according to the time of day.

Intelligent multimedia system, or audiovisual devices for comfort, such as TV, radio, phone etc. Or it can be a system controlled by your voice.

Automatically adjustable temperature and humidity.

Well-equipped security systems, for example alarms and cameras.

Intercom system for communication between rooms in the house.

Other special devices, like a remote-controlled washing machine, oven, vacuum-cleaner or your bed.

6. Charts

This screen shows a chart about advantages and disadvantages.

Despite of the fact that the smart home has many advantages, unfortunately it also has disadvantages.

Living in a smart home is much more convenient and entertaining, isn't it? Besides, you can save a lot of time and energy. As a woman, I can tell you that this is the best point, because of cleaning.

This system also supports the lives of elderly and disabled people. For example: In addition to stairs, an elevator can also be built into the house.

However, it requires a huge investment, which not everyone can afford.

Furthermore, the house is vulnerable to power and Internet outages.

Unfortunately, not every type of house can be transformed into a smart home. An apartment is not the most ideal choice.

In summary how safe it is? I would say that is neither an advantage nor a disadvantage, because there are plenty of cameras and other security devices available, however there is a risk of electronic hacking or even obtaining information for commercial purposes.

This diagram shows how many people have been interested in the smart home since 2004. It reached its peak in 2018, and since then, interest started to decline a little. On this world map, we can see the interest in the topic by countries. I highlighted the top 5 country and Hungary. We are ahead of Sweden and behind Denmark in the ranking.

In conclusion, the concept of smart home is fantastic and modern, but this technology is still in it's infancy.

Before ending my presentation, let me show you a video about a special guest.

That brings me to the end of my presentation. Thank you for your attention! If you have questions I'll do my best to answer them.